



Finite element modelling and validation for breast cancer detection using digital image elasto-tomography

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Abstract

Finite element (FE) models are increasingly used to validate experimental data in breast cancer. This research constructed a biomechanical FE model for breast shaped phantoms used to develop and validate a mechanical vibration based screening system. Such models do not currently exist but would enhance development of this screening technology. Three phantoms were modelled: healthy, with 10 and 20 mm inclusions. The overall goal was to create models with enough accuracy to replace experimental phantoms in providing data to optimize diagnostic algorithms for digital image-based elasto-tomography (DIET) screening technologies. FE model results were validating against experimental DIET phantom data for over 4000 collected points on each model and phantom using cross-correlation coefficients between experimental simulated data and direct comparison. Results showed good to strong correlation ranging from 0.7 to 1.0 in all cases with over 90% having a value over 0.9. Magnitudes for each frame of the dynamic response also matched well, indicating that the material properties and geometry were accurate enough to provide this level of correlation. These results justify the use of FE model generated data for in silico diagnostic algorithm development testing. The overall modelling and validation approach is not overly complex, and thus generalizable to similar problems using mechanical properties of silicone phantoms, and might be extensible to human cases with further work.

Keywords Finite element method · Digital image-based elasto-tomography · Breast cancer · Cross-correlation coefficient

1 Introduction

Breast cancer is the second most common cancer in the world and has the highest cancer mortality rate in women. Presently, one out of every four women carry the risk of breast cancer [1], of which, 6% are below the minimum screening age of

45 years [2]. Early detection via regular screening not only reduces mortality rates, but also decreases overall treatment expenditure [3].

X-ray mammography is the clinical standard breast cancer screening technique. However, due to significant limitations, such as unacceptable rates of false positive and false negative results [4], exposure to ionizing radiation, discomfort and pain due to the compression of the breast [5], it is not recommended for screening. Additionally, younger women under 45 years of age are not eligible for mammography because of their dense breast. Ultrasound was also receiving particular attention. Ultrasound is currently used to differentiate breast masses [6] and guide aspirations and biopsies [7]. Similar to mammography, ultrasound also has limitations. It too requires a well-trained, skilled operator and a follow-up team that includes radiologists [8], and the impact of radiologist experience on diagnostic accuracy [8] means that there is a need for a better screening method, as reviewed in [9]. In particular, there is a need for methods that mitigate these issues.

The digital image elasto-tomography (DIET) concept is based on non-invasive, vibration based analysis of local tissue stiffness elasto-graphic reconstruction [10–15]. The DIET system consists of five digital cameras and synchronized strobe

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lights to capture steady-state surface motion in response to 16–50 Hz low amplitude sinusoidal mechanical actuation applied to the breast [16]. Several images are captured during one cycle of a given actuation frequency at different phases relative to the input motion [17–21]. Surface motions are used to assess local tissue stiffness for diagnosis of stiffer inclusions (e.g. [12, 13]). Recent analysis showing annual mammography down to 40 years of age would save many more lives than current screening [22], which indicates a system like DIET, safely offering screening to any age could improve breast cancer mortality.

DIET was developed using physical silicone phantoms [23, 24]. However, a trial and error experimental approach can be time consuming when developing algorithms to analyse surface motions and create diagnostic metrics, particularly when testing new phantom cases is necessary. An accurate computational model able to repeat the behaviour of these phantoms would enable *in silico* development of analytical diagnostic metrics for DIET and any similar elastography methods for any type of phantom. Similarly, successful computational models and methods for simulating this type of physical phantom would be a first step in translation to modelling human breast tissues for similar research and development goals.

Finite element (FE) modelling enables analysis and evaluation of structures and systems for which there are no theoretical models or solutions because of the complexity of the material properties and boundary conditions. In biomedicine, some applications include guiding clinical biopsies [25] and modelling compression [26]. X-ray mammography to localize mammogram identified inclusions [27, 28], registering X-ray and MR mammography, validating non-rigid registration algorithms [29] and testing reconstruction algorithms for elastography [30]. Thus, with regard to the latter approach, elastography, generally full volume analyses with data from ultrasound or MRI, has a long history but differs from the DIET concept, which utilizes only surface motions and digital imaging. Hence, FE models have a long history in the broad area of breast cancer screening.

The DIET concept is based on mechanical vibrations and its development includes extensive use of silicone phantoms, which can be tested repeatedly in developing hardware, software or diagnostics. Thus, a finite element model of these phantoms that is able to provide the same data as an experimentally tested phantom would enable far more rapid analysis and diagnostic development by replacing experimental tests with much faster and more repeatable finite element analyses. In particular, it is far faster and easier to develop and run a finite element model, of a phantom, in a given configuration with/without inclusions, than it is to develop and fabricate the phantom and then to test it physically. However, while finite elements have been used extensively in areas related to breast cancer, there are no studies on this modelling for

use in mimicking the mechanical properties of breast tissue. The closest prior work is that of Lee et al. and Rajagopal et al. [28, 31] who model the high strain deformation of breast tissue in mammography.

Hence, this research presents the development and validation of finite element model approach to analyse breast-shaped silicone phantoms used to mimic cancerous lesions with and without stiffer inclusions. A novel biomechanically motivated model validation methodology is presented, based on surface displacements of the breast phantom, which is the key diagnostic metric in the DIET concept. In particular, validation focuses on the requirement for this modal to capture dynamic, steady-state response at all phases, rather than just amplitudes, imposed by using such models to develop the DIET concept.

2 Materials and methods

A silicone-based homogenous breast-shaped phantom was developed to perform experimental testing. Phantom dimensions and fabrication procedures were followed from published reports [13, 24], where the 1-mm-thick skin layer was excluded. An example is shown in Fig. 1a.

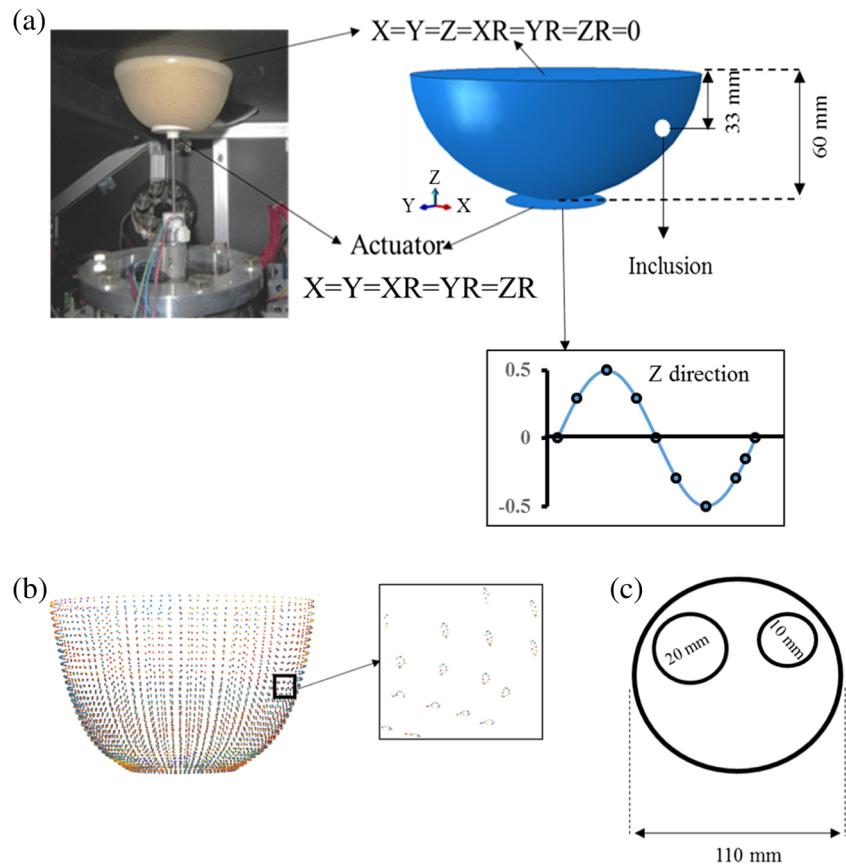
2.1 DIET experimental data

The DIET system images a range of breast shapes and sizes, so the position of the breast and actuator in each image varies between trials. Spherical coordinates (r, θ, φ) are used to estimate a parametric 3D model of the breast surface from frame images, for each of the $k = 10$ frames imaged over one cycle. A grid of reconstructed 3D surface points is then projected onto the breast image. To estimate surface motion of these points, an optical flow algorithm is applied to pairs of frames [32]. Figure 1b shows points on a breast phantom moving in a single loop direction in response to sinusoidal input after $k = 10$ frames in a response cycle. One motion set includes $\sim 15,000 \times 10$ points per phantom or breast [32].

While silicone phantom experiments are important to measure accuracy of the specific modelling features, the silicone phantoms were not intended to perfectly represent the breast. However, the material properties of the silicone used and their fabrication are designed to mimic realistic breast tissues [23, 24]. Silicone with approximately four times greater stiffness than that used to mimic healthy tissue [33] is used to represent stiffer lesions or inclusions, which are located as in Fig. 1c. Overall, these phantoms capture essential shape, as well as healthy and inclusion material properties, to provide a realistic phantom for development.

Figure 1a–c defines the three phantoms, including the no inclusion healthy case, tested. Data from 30 Hz with 1 mm amplitude input motion is considered. In each case, a 2-N

Fig. 1 Total three phantoms were constructed, a healthy, with a 20-mm inclusion and a 10-mm inclusion. **a** Phantom in the DIET prototype. Also shown is the geometry, dimensions, boundary conditions, and loading direction; the top of the phantom is fixed in all translations and rotations direction $X = Y = Z = XR = YR = ZR = 0$ and also actuator displacement is on Z direction and X and Y translations and all three rotations (XR , YR , ZR) are fixed. **b** Points on the breast motion after 10 frames; each frame represents different colour. **c** Summary of two phantoms with a position of 20 and 10 mm inclusions (top view)—two phantoms were created, and in each phantom, 20 and 10 mm inclusions were inserted



preload is also applied, which ensures full contact between the phantom and the actuator.

2.2 Finite element modelling

2.2.1 Geometry

The FE model geometry has the same dimensions as the fabricated phantom shown in Fig. 1a, b. The model assumes a homogenous, isotropic and non-linear material [25]. For two phantoms, spheres of 20 and 10 mm diameter were created and inserted to mimic an inclusion. Their location is also shown in Fig. 1c.

2.2.2 Boundary conditions and input loading

Boundary conditions represent the initial conditions applied to the model. Fixed boundary conditions are applied to nodes on the chest wall. The second boundary condition is at the actuator. The displacement of actuator is strictly vertical, and thus, $X = Y = XR = YR = ZR = 0$ for X and Y translations and all three rotations (XR , YR , ZR).

In simulation, to mimic conditions in the actual DIET system, a vertical, static force of 2 N is applied as a preload on actuator z -axis to ensure contact with the breast is maintained and also this compressive force on the phantom improves the

repeatability of results. A 1-mm peak-to-peak input displacement at 30 Hz is then applied on z -axis.

2.2.3 Mesh convergence

3D solid elements were defined in the breast phantom volume, including for the inclusions where necessary. The challenge is defining the finite-element mesh that best mimics the structure of the breast shaped phantom in simulation, while minimizing model size and complexity. Soft breast tissue is assumed to show incompressible behaviour [34]. Thus, a 3D free mesh was assembled using quadratic tetrahedral 10-node (C3D10M—continuum three-dimensional 10-node modified) elements, which is especially attractive for complex models or curved surfaces [35]. Figure 2a shows a 10-node tetrahedral element.

DIET experimental results provide phantom surface motion displacement information. Mesh convergence was thus assessed based on nodal displacements to determine the mesh resolution yielding an accurate (within 5%) prediction of the measured static deformations for the initial loading conditions. The displacement of the 2 N applied preload was recorded for each node on the surface of the modelled breast phantom. These nodes are expected to compress due to the loading condition imposed. The deformation variation was calculated for each mesh refinement to choose the correct mesh size to validate the breast model. Element size was varied from 2 to 10 mm.

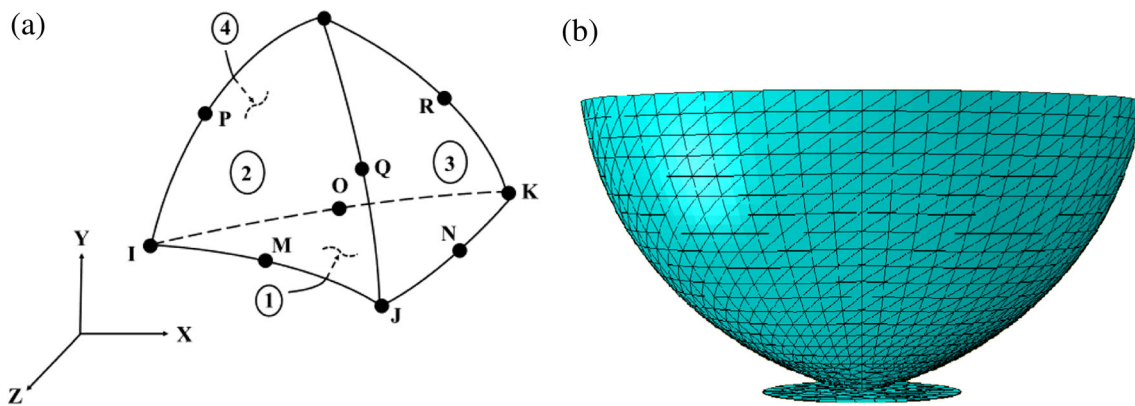


Fig. 2 **a** Tetrahedral element, a 10-node quadratic tetrahedral element, as reproduced from [36]. **b** Geometry with mesh size 4 mm

FE models with 2–4 mm elements had similar results. Thus, FE models with 4 mm elements were used for further analysis. The resulting model has a total of 56,611 nodes and 39,867 elements. Figure 2b shows geometry with mesh size 4 mm.

2.2.4 Material properties

Each element is assigned material properties corresponding to healthy and stiffer inclusion silicone phantom tissues. Hyper-elastic materials are well adapted to simulate soft tissue deformation. Based on previous work, a Neo-Hookean hyper-elastic model is chosen to simulate these breast phantom tissues [31, 37]. Most biological tissues are also considered isotropic, homogenous and incompressible. With these assumptions, it is possible to define the mechanical behaviour of breast phantom tissue using a single elastic modulus, E , with neo-Hookean properties, which is the reduced version of the Mooney Rivlin model [38], with $C_{10} = 0.67$ kPa for fat tissue and $C_{10} = 15.0$ kPa for cancerous tissue [24]. The strain energy density for the Neo-Hookean material is defined:

$$W = C_{10}(I_1 - 3) \quad (1)$$

where I_1 is the strain invariants, and C_{10} is the material constant which is related to shear modulus. For isotropic and incompressible materials, using uniaxial data, the stress and stretch are related by a Neo-Hookean model defined as follows:

$$\sigma = 2C_{10}(\lambda^2 - \lambda^{-1}) \quad (2)$$

where λ is the stretch ratio, and small strain shear modulus μ is given by $2C_{10}$.

2.3 Model validation

In this section, the $k = 10$ frame radial displacement magnitudes from the sinusoidal response are compared first. These absolute values provide a dynamic size validation per frame of the sinusoidal response. They thus assess whether the overall material

properties and model yield the correct magnitude of response. If each frame size is good, the overall trend and sign across the frames of these cycles are validated using a cross-correlation coefficient. The second step assesses the specific trend and profile of the dynamic response. This combined method thus validates first the overall size and second the dynamic trajectory.

2.3.1 Frame-to-frame radial displacement magnitude

DIET captures several images over one sinusoidal response cycle at different phase angles ($0^\circ, 45^\circ, 90^\circ, 135^\circ, 0^\circ, -45^\circ, -90^\circ, -135^\circ, 0^\circ$) relative to the input motion, creating $k = 10$ images per cycle. Over a phantom, one motion set includes $\sim 15,000 \times 10$ surface points. However, the FE model mesh has 4200×10 surface points. For validation, the DIET motion set is reduced from $\sim 15,000 \times 10$ points to 4200×10 FE model mesh points.

Figure 3 explain how the frame-to-frame radial displacement for each of the 4200 points is calculated by Eq. (3):

$$\Delta r_i^k = \sqrt{(x_{(i,k-1)} - x_{(i,k)})^2 + (y_{(i,k-1)} - y_{(i,k)})^2 + (z_{(i,k-1)} - z_{(i,k)})^2} \quad (3)$$

where Δr_i^k is the frame-to-frame radial displacement and $k = 1 \dots 10$ are number of frames and $i = 1 \dots 4200$ are number of

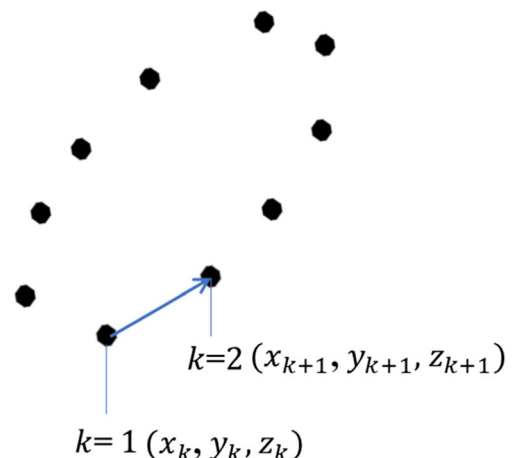


Fig. 3 Frame-to-frame displacement for each of the 4200 points

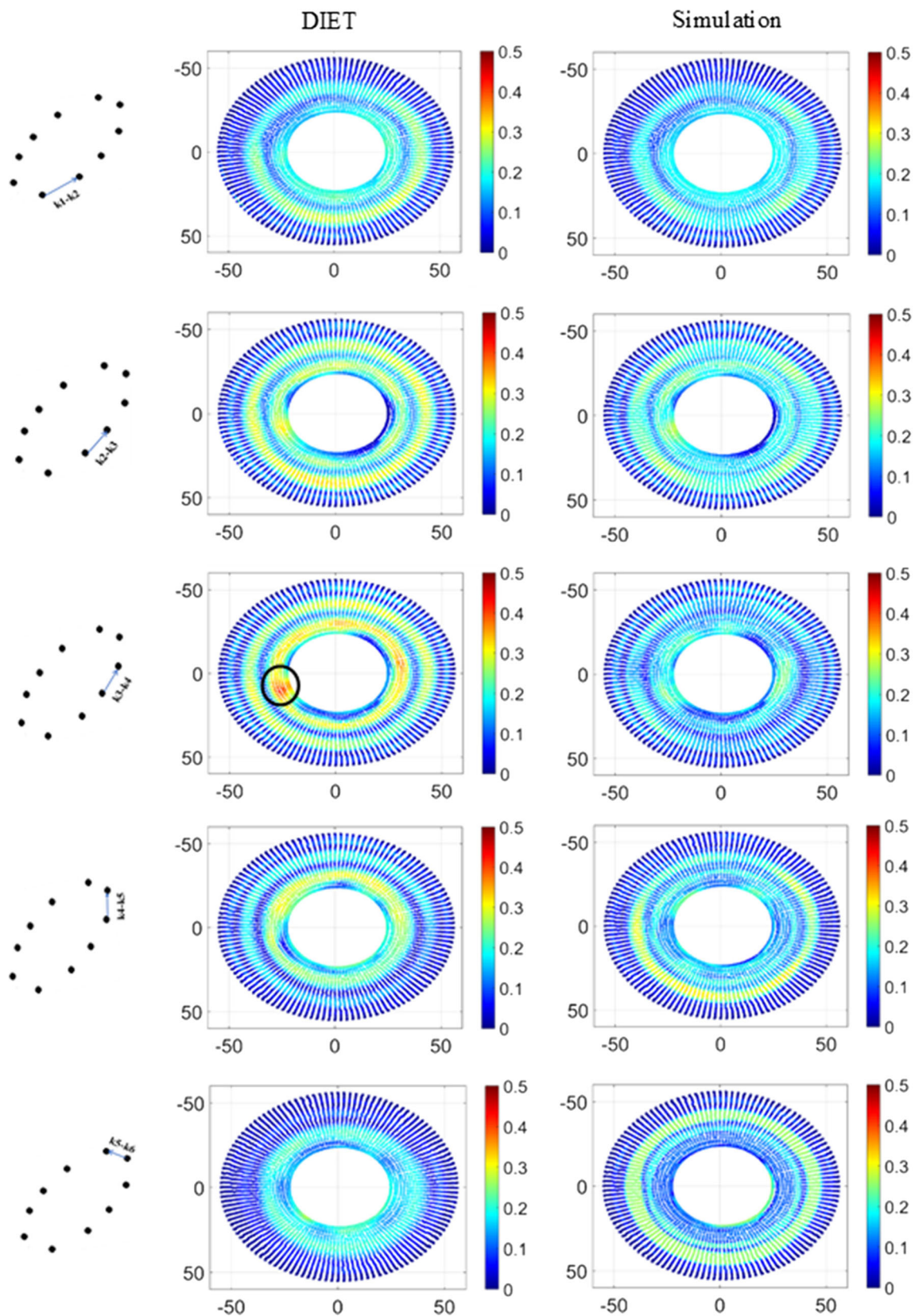


Fig. 4 Displacement distribution (all units in mm) of healthy phantom for experimental DIET (left) and FE model (right) for the $k = 1 \dots 10$ frames captured (top-bottom)—displacement distribution from frames 3 to 4 near

the actuator is uneven, indicated by a black circle, leading to some loss of symmetry

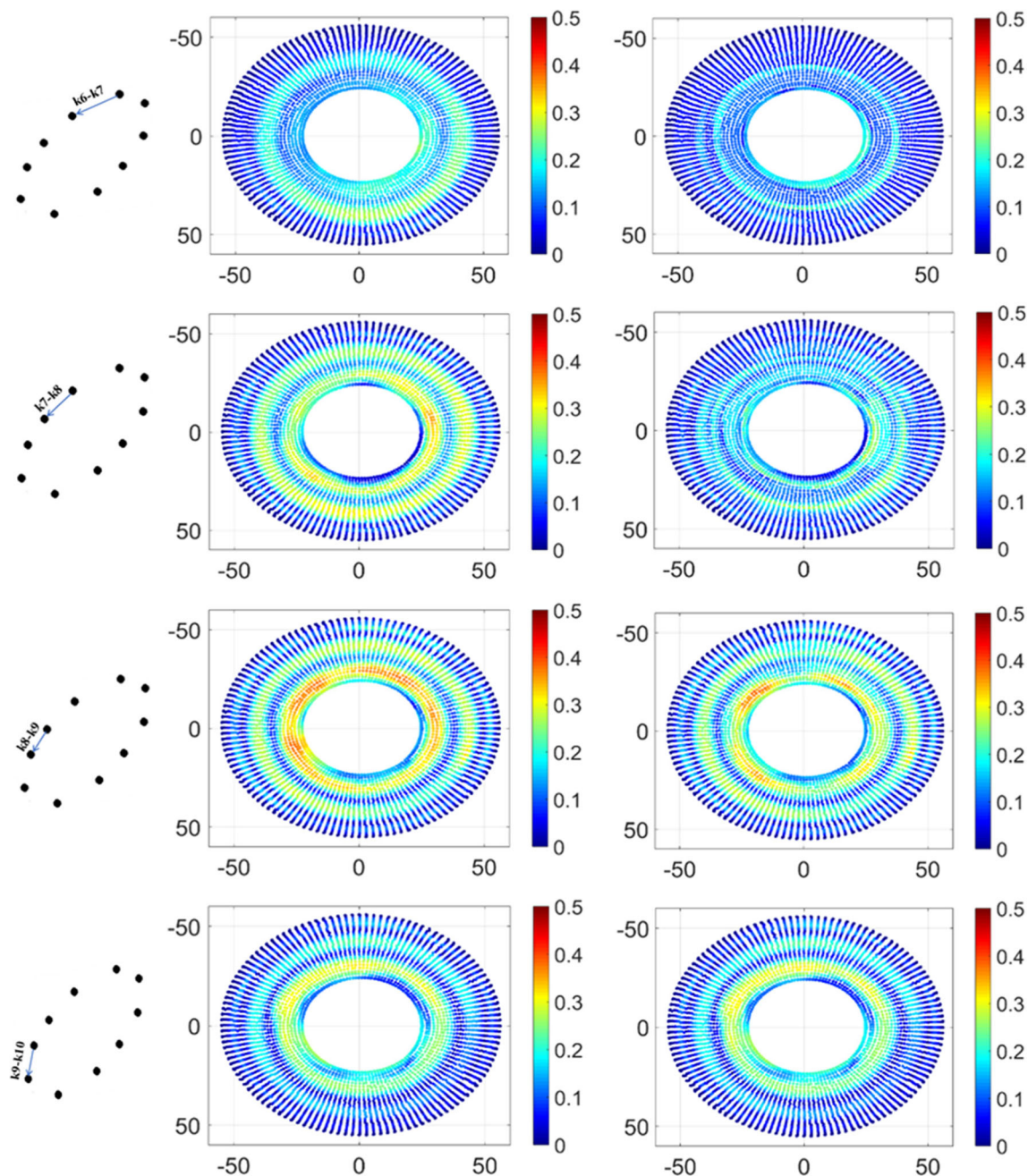


Fig. 4 (continued)

points. The overall average values over all $k = 10$ frames are also compared and defined in Eq. (4)

$$\overline{\Delta r_i} = \frac{1}{10} \sum_{k=1}^{10} \Delta r_i^k \quad (4)$$

2.3.2 Cross-correlation and accuracy of the dynamic trajectory

The cross-correlation coefficient is used to validate the similarity of dynamic trajectory across both data sets. All cross-correlation coefficients range from -1.0 to $+1.0$. A correlation

coefficient of ± 1.0 indicates a perfectly identical or opposite relationship between the two variables, and thus both variables increase or decrease together or oppositely, perfectly. A correlation coefficient of 0.0 indicates zero correlation, or no relationship, between the two variables. The closer a correlation coefficient is to 0.0 , the weaker the relationship and the less able one variable is to define the behaviour of the second.

The vectors $\overrightarrow{\Delta r_i^k}$ for experiment and FE model are compared, for all $i = 1 \dots 4200$ points, and $k = 1 \dots 10$ frames, where the $\overrightarrow{\Delta r}$ vectors are defined for both experiment and FE model from

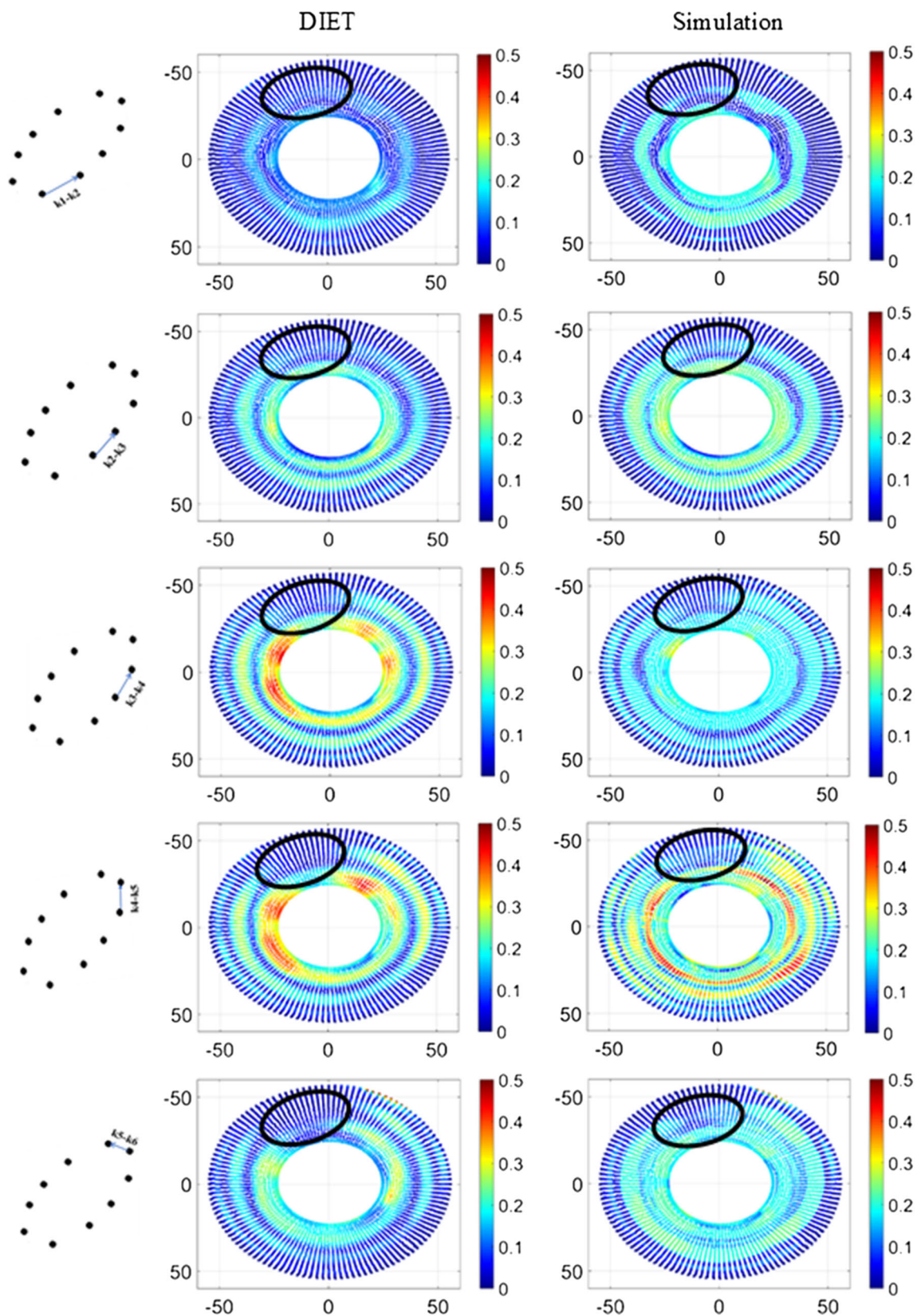


Fig. 5 Displacement distribution (all units in mm) of 20 mm inclusion phantom for experimental DIET (left) and FE model (right) for the $k = 1 \dots 10$ frames captured (top-bottom) position of 20 mm inclusion locate with black circle

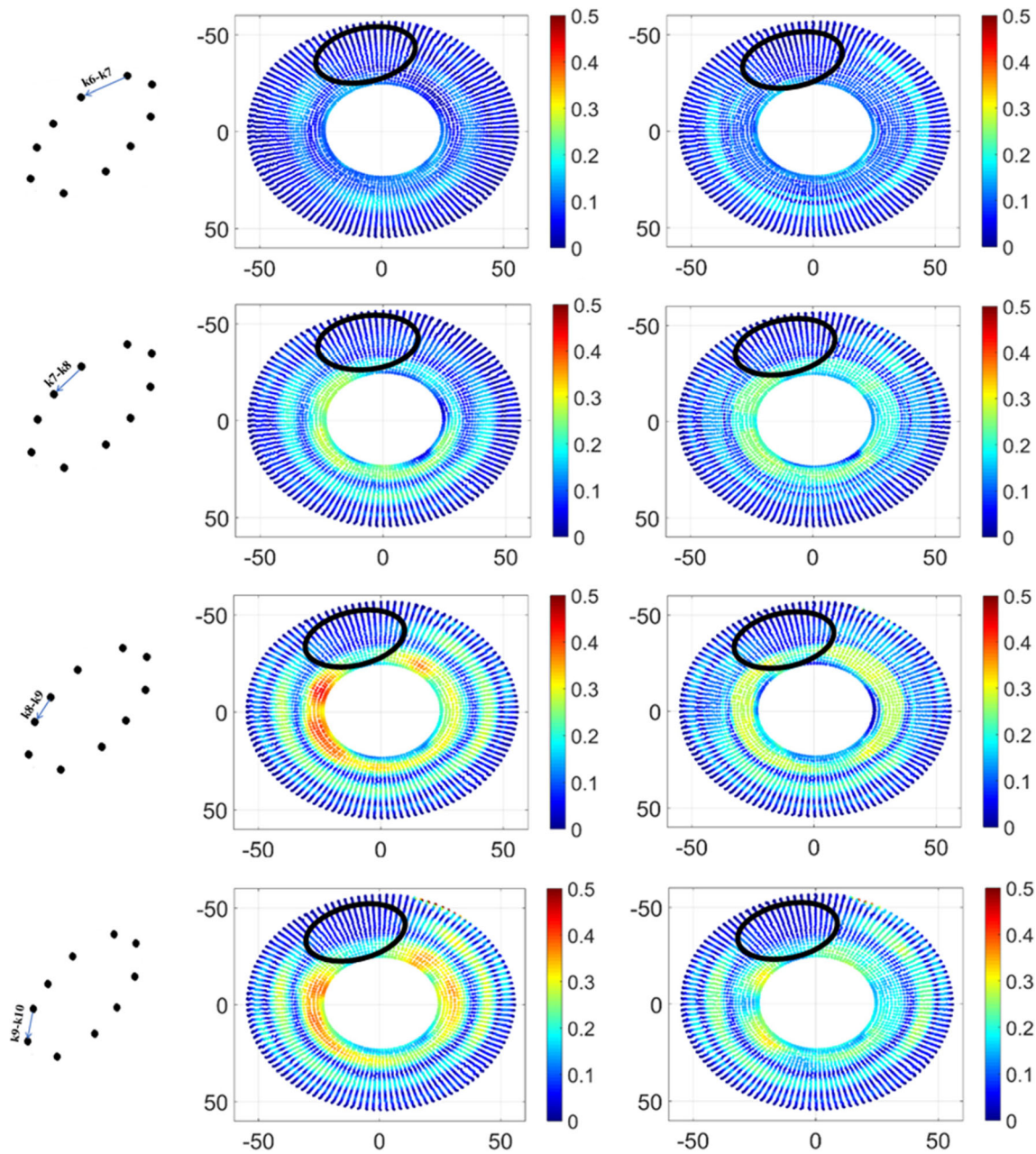


Fig. 5 (continued)

Eq. (3). Thus, $\overrightarrow{\Delta r_i^k}(\text{expor FEM}) = [\Delta r_i^1, \Delta r_i^2 \dots \Delta r_i^{10}]$, where the $\overrightarrow{\Delta r}$ vectors are defined for both experiment and FE model.

The zero lag cross-correlation, x of $\overrightarrow{\Delta r_i^k}(\text{exp})$ and $\Delta r_i^k(\text{FEM})$ used here are defined as follows:

$$x = \frac{(\overrightarrow{\Delta r_i^k}(\text{exp}) \cdot \overrightarrow{\Delta r_i^k}(\text{FEM}))}{\|\Delta r_i^k(\text{exp})\| \|\Delta r_i^k(\text{FEM})\|} \quad (5)$$

where $\overrightarrow{\Delta r_i^k}(\text{exp}) \cdot \overrightarrow{\Delta r_i^k}(\text{FEM})$ is the inner or dot product,

and $\|\Delta r_i^k(\text{exp})\| \|\Delta r_i^k(\text{FEM})\|$ is the z -norm magnitude of the vector.

3 Results

3.1 Frame-to-frame radial displacement magnitude validation

Figures 4, 5 and 6 show frame-to-frame validation results for each of the $k = 1 \dots 10$ FE and DIET radial displacement

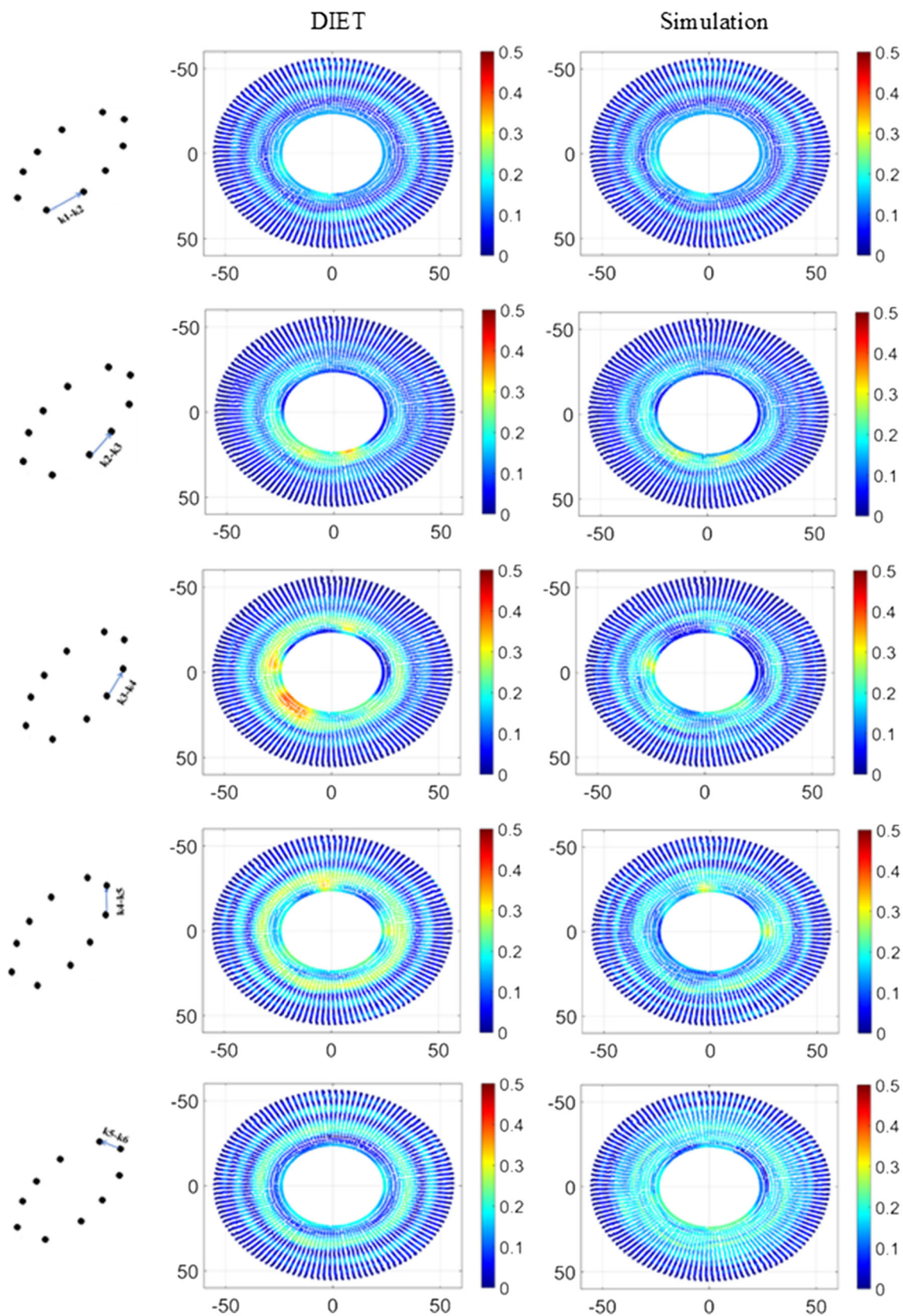


Fig. 6 Displacement distribution (all units in mm) of 10 mm inclusion phantom for experimental DIET (left) and FE model (right) for the $k = 1 \dots 10$ frames captured (top-bottom)

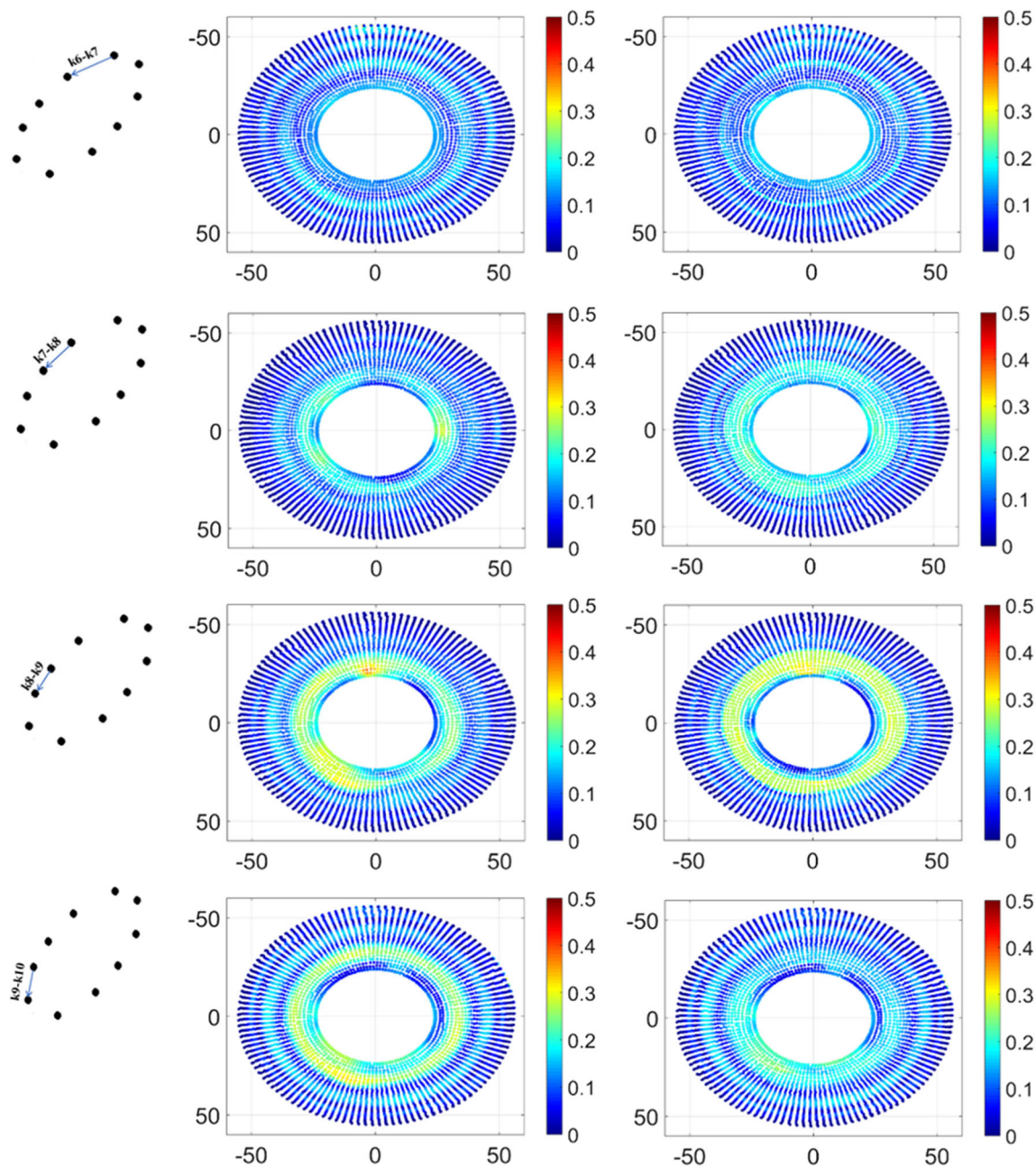


Fig. 6 (continued)

frames (Δr_i^k). The maximum displacement on the colour bars was fixed to 0.5 mm matching the amplitude of the 30 Hz input sine wave. All figures show larger displacements by the actuator and smaller to zero at the chest wall, as expected.

Figure 4 shows the displacement comparison for the healthy phantom. Because the phantom is homogenous, the displacement is evenly distributed. The displacement near the actuator is high due to that input sine wave was applied on the actuator. Additionally, fixed boundary condition was applied on the top of chest wall; therefore, the displacement distribution is zero.

Similarly, Fig. 5 shows the displacement comparison for the phantom with a 20-mm inclusion. Tissue stiffness plays an important role for diagnosis of breast cancers, as inclusions are 4–10 times stiffer than the surrounding breast tissues [39]. The tissue displacements are inversely proportional to the stiffness of the tissue, and thus a stiffer region of tissue exhibits small displacements than a more compliant region [12, 13]. Figure 4 indicates the position of the 20 mm inclusion with a black circle. Again, qualitatively, the match is good, but there are small local differences in magnitude due to imperfect match of phantom and model stiffness or damping or other modelling errors.

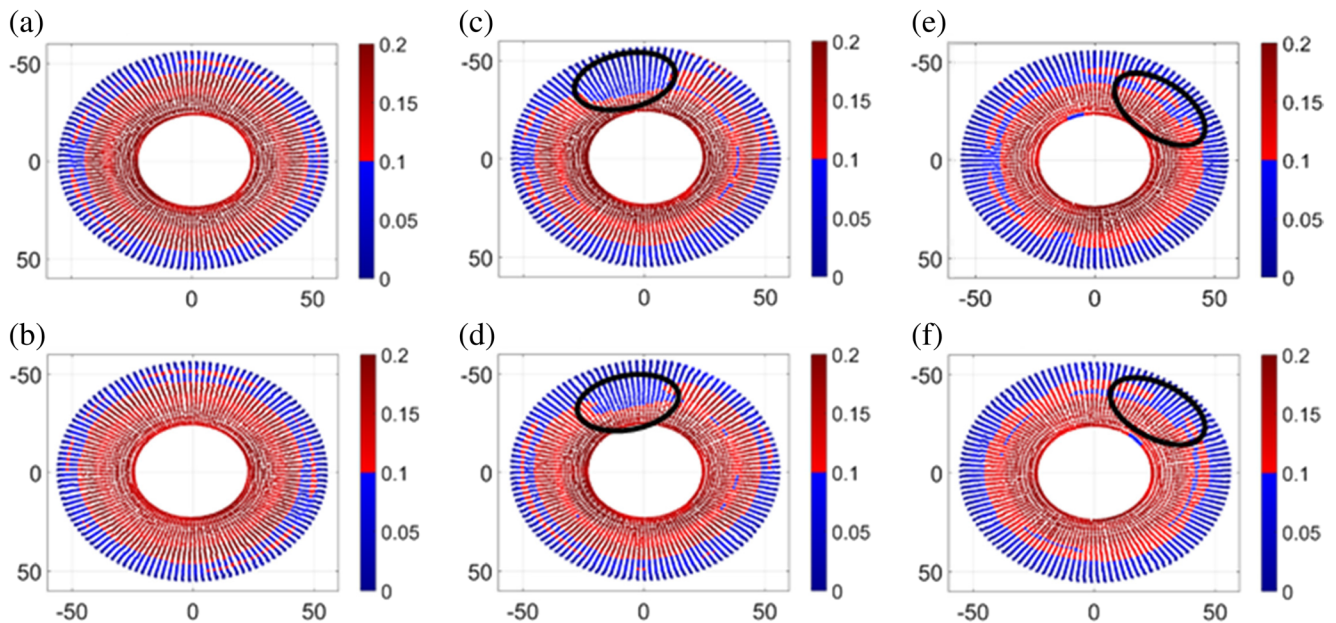


Fig. 7 Average radial displacement distribution (all units in mm) for 30 Hz frequency. **a** DIET healthy phantom. **b** Finite element analysis (FEA) healthy phantom. **c** DIET 20 mm inclusion position locate with

black circle. **d** FEA 20 mm inclusion position locate with black circle. **e** DIET 10 mm inclusion position locate with black circle. **f** FEA 10 mm inclusion position locate with black circle

However, the frame-to-frame magnitudes qualitatively match well.

Figure 6 shows the displacements for the 10 mm inclusion phantom. Because the changes in surface displacement are smaller, it is harder to see a difference between the region where the 10 mm inclusion is located and the surrounding tissues, and thus it cannot be detected in this fashion. As with Figs. 4 and 5, the trends and overall qualitative comparison are adequate.

Figure 7 shows the average radial displacement of each node on the phantom surface ($\Delta \bar{r}_i$) to locate the 20 and 10 mm inclusion with a black circle. Figure 6 shows stronger overall match between DIET experimental and FE model results, as well as in motion amplitudes (average radial displacement), where all $k = 1 \dots 10$ frames are considered. This last result provides further strong qualitative validation but does not provide a specific numerical level for that correlation in matching values. Nor does it provide the necessary trend validation showing how the dynamic response that provides these values in Figs. 4, 5, 6 and 7 is, or is not, similar. This last criterion requires the cross-correlation coefficient value, and quantifies this level of validation explicitly.

3.2 Accuracy of dynamic trajectory using cross-correlation coefficient

The cross-correlation coefficient for all 4200 points overall 10 frames was calculated to analyse discrepancies in the trajectory of the dynamic response between FEA and DIET

experimental results. It measures the degree to which both data sets vary in the same direction or oppositely. It is thus a strong validation for DIET, which uses dynamic response overall frames as a diagnostic. The resulting cumulative distribution functions for each phantom of these values are shown in Fig. 8. Table 1 summarizes these results. Over 90% of points have a cross-correlation coefficient, $\rho \geq 0.9$, indicating very good correlation. Lower values are likely due to the existence of many local effects not taken into account by the more homogenous finite element modelling.

4 Discussion

The overall results in Figs. 4, 5 and 6 matched well. For the healthy phantom in Fig. 4, the displacement is expected to be evenly distributed due to its homogenous, symmetric fabrication. However, due to the 2 N preload deformation, the displacement distribution from frames 3 to 4 near the actuator is uneven, indicated by a black circle, leading to some loss of symmetry. This effect is also seen in Figs. 5 and 6 in other frames for the 20 and 10 mm inclusion phantoms. Otherwise, the overall match between DIET experiments and the FE model is qualitatively good, but these local differences are evident.

The cross-correlation values in Fig. 8 and Table 1 are very good. Thus, while small errors in material properties may cause motion magnitude differences, as seen in Figs. 4, 5 and 6, the overall dynamic response is well matched. Hence, the trajectory across all $k = 1 \dots 10$ frames appears

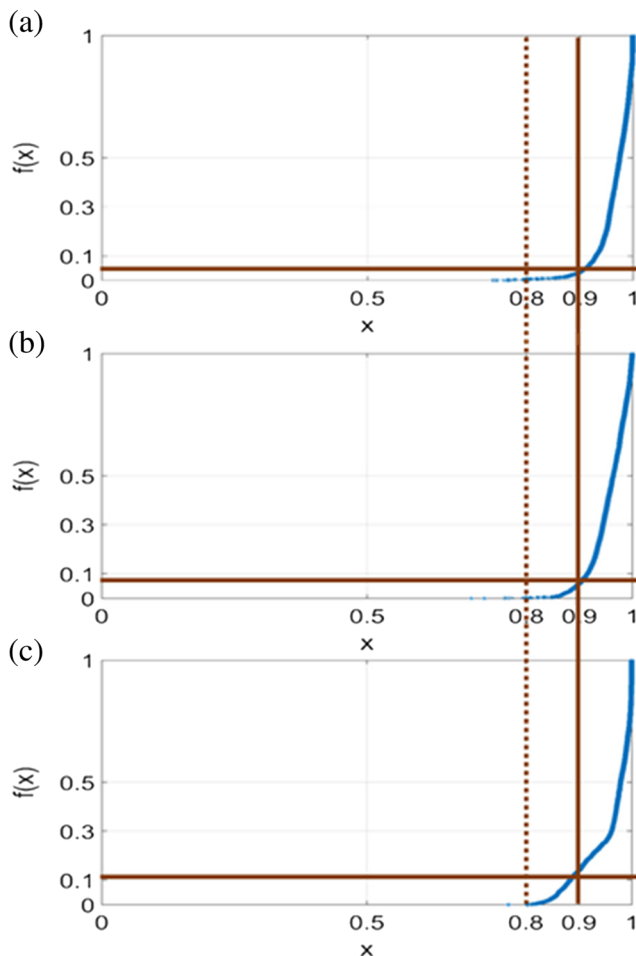


Fig. 8 Empirical distribution ($f(x)$) of cross-correlation coefficient (x) of all points for **a** healthy phantom, **b** 10 mm inclusion and **c** 20 mm inclusion. The solid lines show the $x=0.90$ value and quantify its likelihood. The dashed line shows the location for $x=0.80$ with ~99–100% of the values having $x \geq 0.80$

valid, which is the key aspect in DIET diagnostics, as the changes in this trajectory, in the presence of stiffer inclusions, that is the focal point of the diagnostic approach. Hence, trajectory matching overall $k=1 \dots 10$ frames is critical to successfully using this FE model, or any similar model, in place of phantom experiments given qualitatively good frame-to-frame magnitude matches.

One primary reason of lower values of cross-correlation coefficient is DIET measurement error. Each node on the surface of the breast follows a loop of 10 points, and there is measurement error on every single point, which can degrade perfect correlation. This error is on the area of 6%, down to 2–3 μm [32], and is thus larger for nodes with smaller motions, such as near the chest wall. In addition, the small local discrepancies in Figs. 3, 4 and 5 will yield much lower cross-correlation coefficients.

In addition, the experimental input sine wave is not always perfect, whereas the FE model input is ideal. This issue is another possible cause for the differences seen, particularly in Figs. 4, 5 and 6. Overall, these limitations are effected by the strong results in Figs. 7 and 8, which are central to demonstrating the utility of these models to replace experimental phantom tests in surface motion analysis and diagnostics metric development.

Overall, these results demonstrate the FEA modelling approach presented can predict surface motion of these silicone breast phantoms to these high correlation levels (over 0.90). The results in Fig. 7 show that it captures the changes necessary for developing diagnostic metrics and analyses from the FE model. The homogenous healthy phantom shows slightly better validation than 20 and 10 mm inclusion cases. However, certain assumptions regarding the finite element modelling, such as excluding skin, gravity and body mass, may also have produced some differences particularly in magnitude of response, as seen in the results of Figs. 4, 5 and 6.

A further limitation is the cancerous lesions within the modelled breast phantoms that were connected to healthy tissues without the ability to slide past one another, a perfect connection with perfect, as modelled, stress-strain transfer. Real lesions are strongly connected to tissues, but as in the phantoms modelled here, it is important to note that there may be some forms of spring or other relationships defining how they connect. In this case, the results were strong enough to ensure a good model match. However, going forward, it will be necessary to investigate the effect on displacement from imposing slip

Table 1 Summary of cross-correlation coefficients

Range of cross-correlation coefficient	Number of data points		
	Healthy	10 mm inclusion	20 mm inclusion
0.9–1.0	4055/4200 (96.54%)	3950/4200 (94.04%)	3619/4200 (86.16%)
0.8–0.9	124/4200 (2.95%)	242/4200 (5.76%)	579/4200 (13.78%)
0.7–0.8	21/4200 (0.50%)	7/4200 (0.16%)	2/4200 (0.04%)
< 0.7	0/4200 (0.00%)	1/4200 (0.02%)	0/4200 (0.00%)

boundary conditions between cancerous tissues and healthy tissues.

Another limitation is a Neo-Hookean hyper-elastic model that is chosen to simulate these breast phantom tissues in this study. The phantoms are relatively homogeneous supporting this assumption. However, real breasts will be more heterogeneous even across similar tissue types. Thus, going forward, it would be interesting to assess more detailed mechanics models, such as the Ogden and Mooney Rivlin models, to investigate their effect on surface displacement distributions, and if they can provide a more detailed model in those cases.

Finally, one limitation of this study is the lack of comparison to other studies. The only research in the area of DIET in general has been by the authors group, as it is still a unique approach. In particular, while elastography is a growing field in general, primarily via ultrasound and MRI data [40–42], this mechanical approach in DIET is unique at this time to the best of the authors' knowledge. In particular, the phantoms normally used for development and research in wave based imaging modalities, such as ultrasound, MRI or mammography, are normally defined by their properties in relation to these modalities and input waves. In this case, the phantoms are for DIET defined by their mechanical properties relative to breast tissues [23, 24], which is distinctly unique from the properties normally considered. Thus, the study presented lacks prior comparisons to the best of our knowledge.

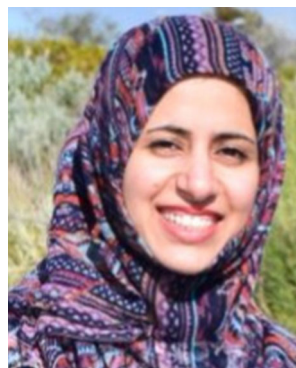
5 Conclusions

The results show an efficient FE breast model for silicone phantoms covering healthy and inclusion cases that provides predictions of the frame-to-frame displacement distribution under dynamic loading conditions with only small differences in magnitude based on high correlation values. These differences are due to a range of unmodelled effects. Any inconsistency between finite element model and DIET experimental data can be improved by reducing experimental errors, as well as improving the material properties used in the model. However, the modelling approach presented can be generalized to other similar cases, and for these biomedical tissue phantoms, it captures all the key dynamics necessary to use them in place of experimental phantom tests in the analysis of surface motions based on the correlation results presented comparing experimental and model results. Thus, it creates a foundation for subsequent development and optimization of DIET diagnostics metrics.

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